MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Surgical Antimicrobial Prophylaxis Recommendations

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II. INTRODUCTION

The use of peri-operative antimicrobials has become an essential component of the standard of care for certain surgical procedures and can result in a reduced risk of post-operative infection when sound and appropriate principles are utilized. However, the benefit of antimicrobial prophylaxis must be weighed against the risks of toxic and allergic reactions, emergence of resistant bacteria, drug interactions, super-infection, and cost.

III. CONSIDERATIONS FOR ANTIMICROBIAL PROPHYLAXIS

Goal: Administer antimicrobial prophylaxis to achieve serum and tissue levels of antimicrobial at the time of incision and for the duration of the operation, that are in excess of the minimum inhibitory concentration (MIC) needed for organisms that may be encountered during the procedure.

a. Antimicrobial prophylaxis should be administered if there is a risk of infection in the absence of a prophylactic agent; clean procedures rarely require prophylaxis unless high risk procedure, including implantation of prosthetic material.
    i. Clean procedures are defined as those with no acute inflammation or transection of gastrointestinal, oropharyngeal, genitourinary, biliary, or respiratory tracts (elective cases, no technique break).
b. The activity of the chosen prophylactic agent(s) should encompass the most common pathogens associated with the surgical procedure and consider local susceptibility data, but need not cover every likely pathogen.
c. The prophylactic agent must be administered in a dose which provides an effective tissue concentration prior to incision / intra-operative bacterial contamination.
    i. In most instances, a single intravenous dose of an antimicrobial agent provides adequate tissue concentrations around the time of anesthesia induction and throughout the operation.
        1. Antimicrobial agent infusion should begin 15-60 minutes before the incision with the exception of vancomycin, levofloxacin, ciprofloxacin, gentamicin, azithromycin and fluconazole. These infusions should begin 45-90 minutes before the incision and infused over 60-120 minutes as indicated for adults and pediatrics (See following tables**).**
        2. In adult patients, cefazolin (2 g if < 120 kg, 3 g if ≥ 120 kg) and vancomycin (1 g if < 80 kg, 1.5 g if ≥ 80 kg) dosing is based on weight. Adult patients < 50 kg should refer to Patients < 50 kg Adult dosing guidance. Dosing recommendations for dosing. Weight-based dosing is recommended for all antibiotics in patients < 50 kg.
        3. Infusion duration and time to redosing for recommended prophylactic antimicrobials are summarized for adults and pediatrics.
        4. **All prophylactic antimicrobials should be discontinued after the intra-operative period, unless otherwise specified.**
            a. Data have not supported subsequent doses after surgical closure and may increase the risk of *Clostridium difficile* and antimicrobial resistance.
            b. A longer duration of antimicrobials may be indicated, if concomitant infection is present at the time of surgery.

BREAST AND AXILLARY PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Excisional biopsies	None	Not recommended	Not recommended
Wire Localized Breast Biopsy, Re-excision lumpectomy, Sentinel (SLN) alone, Lumpectomy & SLN, Axillary Lymph Node Dissection, Mastectomy (Total or Modified Radical)	*S. aureus* *S. epidermidis*	Adult: **Cefazolin** OR **Cefuroxime**	Adult: **Clindamycin** OR **Daptomycin**

CARDIOTHORACIC PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Cardiac surgery with or without implants • Aortic grafts • Prosthetic valves • TAVR **Deep Hypothermic Circulatory Arrest (DHCA)**	*S. aureus* *S. epidermidis* gram-negative bacilli	Adults: **Daptomycin** + **Cefuroxime** All Daptomycin dosing (based on actual body weight) • Weight < 125kg = 500mg • Weight ≥ 125kg = 750mg **Post op daptomycin** dosing: • CrCl < 30 mL/min: no post op doses necessary • CrCl ≥ 30 mL/min: x1 dose (24hrs from pre op dose) **Post op cefuroxime** dosing: • CrCl > 30 mL/min: 1.5 g IV q8h x5 dose • CrCl 10-29 mL/min: 1.5 g IV q12h x3 doses • CrCl < 10 mL/min: 1.5 g IV q24h x1 dose	Adults: **Daptomycin** + **Gentamicin** Gentamicin redosing not recommended given decreased excretion following cardiopulmonary bypass *Alternative if* SCr ≥ 2 mg/dL or CrCl < 40 mL/min: **Daptomycin** + **Levofloxacin** Levofloxacin redosing not indicated given long half-life, especially with renal impairment *Alternative to daptomycin* **Vancomycin**

CARDIOTHORACIC PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Cardiac surgery without implants • CABG alone	*S. aureus* *S. epidermidis*	Adults: **Daptomycin** + **Cefuroxime** All Daptomycin dosing (based on actual body weight) • Weight < 125kg = 500mg • Weight ≥ 125kg = 750mg **Post op daptomycin** dosing: based on CrCl: • CrCl < 30 mL/min = no post op doses necessary • CrCl ≥ 30 mL/min = x1 dose (24hrs from pre op dose) **Post op Cefuroxime** dosing: based on CrCl: • CrCl > 30 mL/min: 1.5 g IV q8h x5 doses • CrCl 10-29 mL/min: 1.5 g IV q12h x3 doses • CrCl < 10 mL/min: 1.5 g IV q24h x1 dose	Adults: **Daptomycin** + **Gentamicin** Gentamicin redosing not recommended given decreased excretion following cardiopulmonary bypass *Alternative to gentamicin if SCr ≥ 2 mg/dL or CrCl < 40 mL/min:* **Daptomycin** + **Levofloxacin** Levofloxacin redosing not indicated given long half-life, especially with renal impairment *Alternative to daptomycin* **Vancomycin** *In case of delayed chest closure, total number of doses should be adjusted to reflect 48h from time of chest closure

CARDIOTHORACIC PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Pacemaker or AICD placement or revision	*S. aureus* *S. epidermidis*	Adults: **Cefazolin** *History of MRSA infection or colonization:* + Daptomycin Continue post-op for 24 hours	Adult: **Daptomycin** *Alternative to daptomycin:* **Vancomycin**

CARDIOTHORACIC PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Heart transplant	*S. aureus* *S. epidermidis* gram-negative bacilli	Adults: **Daptomycin** + **Cefuroxime** All daptomycin dosing (based on actual body weight) • Weight < 125kg = 500mg • Weight ≥ 125kg = 750mg Post op daptomycin dosing based on CrCl: • CrCl < 30 mL/min= no post op doses necessary • CrCl > 30 ml/min= x1 dose (24hrs from pre op dose) Post op cefuroxime dosing based on CrCl: • CrCl > 30 mL/min: 1.5 g IV q8h x5 doses • CrCl 10-29 mL/min: 1.5 g IV q12h x3 doses • CrCl < 10 mL/min: 1.5 g IV q24h x1 dose	Adults: **Daptomycin** + **Levofloxacin** **Levofloxacin** dosing modification: CrCl > 50 mL/min: **Levofloxacin** 500 mg IV q24h x2 doses CrCl ≤ 50 mL/min: **Levofloxacin** 500 mg IV q48h x1 dose *Alternative to daptomycin:* **Vancomycin** *In case of delayed chest closure, total number of doses should be adjusted to reflect 48h from time of chest closure If definitive cultures are available, continue antibiotics and tailor regimen

CARDIOTHORACIC PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Left Ventricular Assist Device (LVAD)	*S. aureus* *S. epidermidis* *Candida* spp. enteric gram-negatives	Adults: **Daptomycin** **+ Cefuroxime** All daptomycin dosing (based on actual body weight) Weight < 125kg = 500mg Weight ≥ 125kg = 750mg Post op daptomycin dosing based on CrCl: • CrCl < 30 mL/min = no post op doses necessary • CrCl ≥ 30 mL/min = 1x dose (24hrs from pre op dose) Start **rifampin** 600 mg PO/IV q24h, **levofloxacin** 500 mg PO/IV q24h and **fluconazole** 400 mg PO/IV q24h post-procedure and continue for 48 hours from OR or from chest closure in case of delayed chest closure*.	Adults: **Daptomycin** + **Levofloxacin** Start **rifampin** 600 mg PO/IV q24h, **levofloxacin** 500 mg PO/IV q24h and **fluconazole** 400 mg PO/IV q24h post-procedure and continue for 48 hours from OR or from chest closure in case of delayed chest closure*. *In case of delayed chest closure, total number of doses should be adjusted to reflect 48h from time of chest closure If allergies or intolerances to vancomycin, rifampin or fluconazole, consultation with Infectious Diseases is recommended.

GASTROINTESTINAL PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Appendectomy (non-perforated)	Enteric gram-negative bacilli Enterococci anaerobes (Bacteroides spp., Clostridia) S. *aureus*	Adults: **Cefazolin** + **Metronidazole** OR **Cefoxitin**	Adults: **Clindamycin** + **Levofloxacin** OR **Clindamycin** + **Aztreonam** *Reserve aminoglycoside-based regimens for patients with intolerance to alternative recommendations or history of documented multi-drug resistant pathogen. If risk factors for acute renal failure present, avoid aminoglycosides if alternative options available:* **Clindamycin** + **Gentamicin**
Colon and anorectal procedures NOT requiring antibiotic prophylaxis: • Evaluation under anesthesia, fulguration of warts, high resolution anoscopy, dilation of stricture, anal biopsy • EUA for fistula placement of seton • Fistulotomy, simple (NOT a Surgisis plug or advancement flap), • Transanal resection of fibroepithelial or pedunculated polyp	None	Prophylaxis Not Recommended	Prophylaxis Not Recommended

GASTROINTESTINAL PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Splenectomy	*S. aureus* *S. epidermidis*	Adults: **Cefazolin** OR **Cefuroxime**	Adults: **Daptomycin**
Colorectal Procedures Not limited to, but including the following: • Colon & rectal resection • High, complex fistula by Surgisis plug or advancement flap • Lateral sphincterotomy for anal fissure • Hemorrhoidectomy • Transanal resection for sessile polyp, villous adenoma, possible T1 malignancy • Rectal prolapse procedure (Altmeier or abdominal rectosigmoid resection/rectopexy) *(Some procedures may be included in SCIP, and appropriate antibiotic selection is linked to hospital reimbursement)*	Enteric gram-negative bacilli Enterococci anaerobes (Bacteroides spp., Clostridia) *S. aureus*	Adults: **Cefazolin** + **Metronidazole** OR **Cefoxitin** Optional oral regimens in combination with IV therapy: **Neomycin** 1,000 mg PO + **Erythromycin** base 1,000 mg PO; give at 19, 18, and 9h before surgery OR **Neomycin** 1,000 mg PO + **Metronidazole** 500 mg PO; give at 19, 18, and 9 h before surgery	Adults: **Clindamycin** + **Levofloxacin** OR **Clindamycin** + **Aztreonam** *Reserve aminoglycoside-based regimens for patients with intolerance to alternative recommendations or history of documented multi-drug resistant pathogen. If risk factors for acute renal failure present, avoid aminoglycosides if alternative options available:* **Clindamycin** + **Gentamicin**

GASTROINTESTINAL PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Esophageal, gastric surgery, G-tube; peg tube (See small bowel for gastric bypass and gastrectomy) ADULTS - Prophylaxis recommended for high risk patients, including diabetes, morbid obesity, cancer, gastric bleeding, gastric outlet obstruction, gastroduodenal perforation, esophageal obstruction, decreased gastric acidity or gastrointestinal motility, morbid obesity	Upper airway flora (aerobic, anaerobic Streptococci) *S. aureus* more rarely aerobic gram-negative bacilli	Adults: **Cefazolin**	Adults: **Daptomycin**
Hepatic, biliary tract, pancreatic including cholecystectomy and gallbladder procedures, cystgastrostomy (see separate section for Pancreaticoduodenectomy) (excluding low-risk laparoscopic cholecystectomy*) *ADULTS – Prophylaxis recommended for high risk patients only, including age > 70 years, non-functioning gall bladder, emergency procedures, diabetes, acute cholecystitis, obstructive jaundice or common duct stones	Enteric gram-negative bacilli (e.g., *E. coli*, *Klebsiella*) Enterococci *S. aureus* anaerobes (*Bacteroides spp.*, *Clostridia*) common with stents, biliary obstruction	Adults: **Cefuroxime** + **Metronidazole** OR **Cefoxitin**	Adults: **Clindamycin** + **Levofloxacin** OR **Clindamycin** + **Aztreonam** *Reserve aminoglycoside-based regimens for patients with intolerance to alternative recommendations or history of documented multi-drug resistant pathogen. If risk factors for acute renal failure present, avoid aminoglycosides if alternative options available:* **Clindamycin** + **Gentamicin**

GASTROINTESTINAL PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Pancreaticoduodenectomy	Enteric gram-negative bacilli (e.g., *E. coli*, *Klebsiella*) Enterococci *S. aureus* anaerobes (*Bacteroides spp.*, *Clostridia*) common with stents, biliary obstruction	Adults **Piperacillin-tazobactam**	Adults **Cefoxitin** **+ Daptomycin** OR **Daptomycin** **+ Aztreonam** **+ Metronidazole**
Small bowel, gastric bypass, gastrectomy	Enteric gram-negative bacilli Enterococci anaerobes (Bacteroides spp., Clostridia) *S. aureus*	Adults: **Cefazolin** + **Metronidazole** OR **Cefoxitin**	Adults: **Clindamycin** + **Levofloxacin** OR **Clindamycin** + **Aztreonam** *Reserve aminoglycoside-based regimens for patients with intolerance to alternative recommendations or history of documented multi-drug resistant pathogen. If risk factors for acute renal failure present, avoid aminoglycosides if alternative options available:* **Clindamycin** + **Gentamicin**

GASTROINTESTINAL PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Hernia repair (hernioplasty-prosthetic mesh repair of hernia; herniorrhaphy-suture repair of hernia)	*S. aureus* *S. epidermidis*	Adults: **Cefazolin**	Adults: **Daptomycin**

GENITOURINARY PROCEDURES

Prophylaxis of genitourinary procedures that enter the urinary tract should include coverage of likely pathogens as well as organisms from the most recent urologic source (urine culture, stone culture, etc) within the preceding year, regardless of prior antimicrobial treatment. If the most recent positive urologic source culture was > 1 year ago, standard prophylaxis should be utilized

Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Urinary Tract Instrumentation*			
Consider Prophylaxis ONLY in patients with risk factors: Cystography, urodynamic study, simple cystourethroscopy, shock wave lithotripsy Risk Factors Include • Poor functional status/frailty • anatomic anomalies of urinary tract • chronic steroid use • immunocompromising condition or recent systemic chemotherapy	Enteric gram-negative bacilli	Adults: **TMP-SMX (Bactrim®)** 1 DS PO, ideally 1-4 hrs prior OR **Amoxicillin-clavulanate** 875 mg PO, ideally 2-4 hrs prior OR **Cefazolin**	Adults: **Gentamicin** 5 mg/kg IV x1 OR 120 mg IM OR **Ciprofloxacin** 500 mg PO 1-2 hrs prior or 400 mg IV
Prophylaxis recommended for: Ureteroscopy ± stent placement Cystourethroscopy with manipulation including: • transurethral resection of bladder tumor and prostate, • any biopsy, resection, fulguration, foreign body removal, urethral dilation or urethrotomy • any ureteral instrumentation including catheterization or stent placement/removal • submucousal injection (e.g., Botulinum toxin) *The following procedures do not require infusion of IV antibiotics timed for 15-30 min prior, but can be given directly before the procedure, as they are intended for prevention for post-procedural UTI: bladder biopsy; botox injection; cystolithalopaxy; cystoscopy with fulguration, microplastique, bladder neck incision or retrograde pyelogram; hydrodistention; ureteral stent placement and removal; ureteroscopy.	*E. coli* *Proteus* spp. *Klebsiella* spp.	Adults: **Cefazolin** 2 g IV/IM; 3 g IV/IM if ≥ 120 kg	Adults: **Gentamicin** 5 mg/kg IV x1 OR 120 mg IM OR **TMP-SMX** (Bactrim®) 1 DS PO x1 OR **Ciprofloxacin** 500 mg PO or 400 mg IV

GENITOURINARY PROCEDURES

Prophylaxis of genitourinary procedures that enter the urinary tract should include coverage of likely pathogens as well as organisms from the most recent urologic source (urine culture, stone culture, etc) within the preceding year, regardless of prior antimicrobial treatment. If the most recent positive urologic source culture was > 1 year ago, standard prophylaxis should be utilized

Percutaneous nephrolithotomy (PCNL) High risk features include: • Positive urine culture within 2-4 weeks • Residual stone with prior positive stone culture • Current indwelling ureteral stent or nephrostomy tube • Severe hydronephrosis • Continuous intermittent catheterization • Renal transplant or other severe immunocompromising condition • Neurogenic bladder (with or without urinary diversion) • Urinary Diversion • Chronic indwelling catheters (Foley or SP tube)	*S. aureus* *S. epidermidis* enteric gram-negative bacilli	Adults w/o high risk features No antibiotics in days preceding PCNL **Cefazolin** prior to procedure Antibiotics should not be continued > 24 hours unless there is concern for post-procedural sepsis Adults w/high risk features: Recommend early ID consultation in anticipation of PCNL Tailored oral antibiotics 3-5 days prior to PCNL, discuss w/ID if no oral option available Tailored IV antibiotic prophylaxis directly prior to procedure Antibiotics only to continue > 24 hrs after procedure for sepsis/complicated UTI due to residual infected stone	Adults: **Daptomycin** + **Gentamicin** If patient allergic to gentamicin or has SCr > 1.5 mg/mL, use **Aztreonam** instead of gentamicin
PROSTATE BIOPSY			
Prostate brachytherapy	*S. aureus* *S. epidermidis* *Streptococci spp.*	Adults: **Cefazolin**	Adults: **Daptomycin**

GENITOURINARY PROCEDURES **Prophylaxis of genitourinary procedures that enter the urinary tract should include coverage of likely pathogens as well as organisms from the most recent urologic source (urine culture, stone culture, etc) within the preceding year, regardless of prior antimicrobial treatment. If the most recent positive urologic source culture was > 1 year ago, standard prophylaxis should be utilized**			
Transperineal prostate biopsy *Routine use of antibiotic prophylaxis is not recommended but can be used in select patients deemed at high risk of infectious complications*	None	Adults: **Cefazolin** OR **Cephalexin** 1000 mg PO 1 hour prior to procedure	Adults: **Doxycycline** 100 mg PO 1 hour prior to the procedure

GENITOURINARY PROCEDURES

Prophylaxis of genitourinary procedures that enter the urinary tract should include coverage of likely pathogens as well as organisms from the most recent urologic source (urine culture, stone culture, etc) within the preceding year, regardless of prior antimicrobial treatment. If the most recent positive urologic source culture was > 1 year ago, standard prophylaxis should be utilized

Transrectal prostate biopsy	*E. coli* *Proteus* spp. *Klebsiella* spp.	Adults: Rectal Swab Performed: *Ciprofloxacin sensitive:* **Ciprofloxacin** 500 mg PO 1 hour prior to procedure and 500 mg PO 12 hours after the first dose OR **Levofloxacin** 750 mg PO 1 hour prior to procedure *TMP-SMX (Bactrim™) sensitive:* **TMP-SMX** 1 DS PO 1 hour before procedure and 1 DS 12 hours after the first dose *Ciprofloxacin resistant and TMP-SMX resistant:* **Cefazolin** 2 g IM; 3 g IM if ≥ 120 kg *Ciprofloxacin, TMP-SMX, and cefazolin resistant:* **Ceftriaxone** 1 g IM No Rectal Swab Performed: **Gentamicin** 120 mg IM + **Ciprofloxacin** 500 mg PO 1 hour prior to procedure and 500 mg PO 12 hours after the first dose OR **Gentamicin** 120 mg IM + **Levofloxacin** 750 mg PO 1 hour prior to procedure	Adults: Rectal Swab Performed: *Allergic or resistant to ceftriaxone, cefazolin, ciprofloxacin, & TMP-SMX:* **Gentamicin** 120 mg IM or 5 mg/kg IV x1 (ideal body weight) If isolated pathogen is resistant to ciprofloxacin, gentamicin, cefazolin, and ceftriaxone then antimicrobials should be chosen based on organism susceptibilities No Rectal Swab Performed: **Cefazolin** 2 g IM; 3 g IM if ≥ 120 kg + **Ciprofloxacin** 500 mg PO 1 hour prior to procedure and 500 mg PO 12 hours after the first dose OR **Cefazolin** 2 g IM; 3 g IM if ≥ 120 kg + **Levofloxacin** 750 mg PO 1 hour prior to procedure

GENITOURINARY PROCEDURES

Prophylaxis of genitourinary procedures that enter the urinary tract should include coverage of likely pathogens as well as organisms from the most recent urologic source (urine culture, stone culture, etc) within the preceding year, regardless of prior antimicrobial treatment. If the most recent positive urologic source culture was > 1 year ago, standard prophylaxis should be utilized

Adrenalectomy, Retroperitoneal/pelvic lymphadenectomy without entering the urinary tract	*S. aureus* *S. epidermidis* *Streptococci spp.*	Adults: **Cefazolin**	Adults: **Daptomycin**
Circumcision	Staphylococci	Adults: None unless diabetes mellitus or other risk factors; then **Cefazolin**	Adults: **Daptomycin**
Genitourinary procedures involving small or large intestine Including urinary diversions, cystectomy with small bowel conduit, uretero-pelvic junction repair, colon conduits, radical cystectomy neobladder/ileostomy, etc	*S. aureus* *S. epidermidis* Streptococci Enteric gram-negative bacilli	Adults: **Cefuroxime** + **Metronidazole** OR **Cefoxitin** Optional oral antimicrobials in combination with above prophylaxis recommendations: **neomycin sulfate** + **erythromycin** base OR **neomycin sulfate** + **metronidazole**	Adults: **Levofloxacin** 500 mg IV/PO + **Metronidazole** If history of MRSA infection or colonization: **Daptomycin** + **Gentamicin** Optional oral antimicrobials in combination with above prophylaxis recommendations: **neomycin sulfate** + **erythromycin** base OR **neomycin sulfate** + **metronidazole**

GENITOURINARY PROCEDURES

Prophylaxis of genitourinary procedures that enter the urinary tract should include coverage of likely pathogens as well as organisms from the most recent urologic source (urine culture, stone culture, etc) within the preceding year, regardless of prior antimicrobial treatment. If the most recent positive urologic source culture was > 1 year ago, standard prophylaxis should be utilized

Penile prosthesis insertion, removal, & revision	*Staphylococci spp.* Enteric gram-negative bacilli	Adults: Cefazolin + Gentamicin + Fluconazole OR Daptomycin + Gentamicin + Fluconazole Antibiotics should not be continued > 24 hours post-operatively unless there is concern for sepsis	Adults: *History of MRSA infection or colonization AND SCr > 1.5 mg/dL:* Daptomycin + Aztreonam + Fluconazole
Testicluar implants	*Staphylocci spp.* *Enteric gram-negative bacilli*	Adults: Cefazolin + Gentamicin OR Daptomycin + Gentamicin	Adults: *History of MRSA infection or colonization AND SCr > 1.5 mg/dL:* Daptomycin + Aztreonam
Implanted prosthetic devices Artificial urinary sphincter and sacral nerve stimulators	Staphylococci enteric gram-negative bacilli	Adults: Cefazolin + Gentamicin OR Daptomycin + Gentamicin	Adults: Daptomycin + Aztreonam

GENITOURINARY PROCEDURES

Prophylaxis of genitourinary procedures that enter the urinary tract should include coverage of likely pathogens as well as organisms from the most recent urologic source (urine culture, stone culture, etc) within the preceding year, regardless of prior antimicrobial treatment. If the most recent positive urologic source culture was > 1 year ago, standard prophylaxis should be utilized

Vaginal urologic surgery Including urethral sling, fistulae repair, urethral diverticulectomy, urethropexy **Open or laparoscopic procedure involving entry into the urinary tract** Including urethroplasty; stricture repair including urethrectomy. Nephrectomy, partial or otherwise, urethrectomy, pyeloplasty, radical prostatectomy, partial cystectomy	Enteric gram-negative bacilli. *S. aureus* *S. epidermidis* *Streptococci spp.* (vaginal surgery mostly)	Adults: **Cefazolin**	Adults: **Levofloxacin** 500 mg IV/PO If history of MRSA infection or colonization: **Daptomycin** **+ Aztreonam**
Vasectomy	*Staphylococci spp.* Enteric gram-negative bacilli	NOTE: Some experts do not recommend prophylaxis for all vasectomy cases. Prophylaxis may be considered based on complexity of procedure and comorbidities of patient. Adults: **Cefazolin**	Adults: *History of MRSA infection or colonization:* **Daptomycin** **+ Aztreonam**
Inguinal and scrotal cases Including radical orchiectomy, reversals, variocelectomy, hydrocelectomy	*Staphylococci spp.* Enteric gram-negative bacilli	Adults: **Cefazolin**	Adults: *History of MRSA infection or colonization:* **Daptomycin** **+ Aztreonam**
Medoidioplasty	*S.aureus* *Streptococcus*	Adults: **Cefazolin**	Adults: **Daptomycin**

HEAD AND NECK PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Clean, non-contaminated procedures (i.e., thyroidectomy, lymph node excision)	None	Not recommended Select clean non-contaminated procedures may confer higher risk for infection and warrant prophylaxis (such as Cefazolin) per surgeon discretion.	Not recommended
Clean contaminated head and neck surgery (incision through oral, pharyngeal, or nasal mucosa)[Ω]	Oral anaerobes enteric gram-negative bacilli *S. aureus* *S. epidermidis* viridans streptococci	Adults: **Ampicillin-sulbactam**	Adults: **Cefazolin** + **Metronidazole** OR **Levofloxacin** + **Metronidazole**
Clean procedure with insertion of a prosthesis (including BAHA hearing device)	*S. aureus* *S. epidermidis*	Adults: **Cefazolin**	Adults: **Clindamycin** OR **Vancomycin**
Skull base, lateral or posterior approach procedures (including cochlear implants)	*S. aureus* *S. epidermidis* *S. pneumoniae*	Adults: **Cefuroxime** OR **Ampicillin-sulbactam**	Adults: **Clindamycin**

HEAD AND NECK PROCEDURES			
Skull base, anterior approach including transphenoidal surgery for pituitary tumors	*S. aureus* *S. epidermidis* gram-negative bacilli	Adults: **Ampicillin-sulbactam** If history of MRSA infection or colonization: + **Vancomycin** No post-operative prophylaxis for routine pituitary adenoma surgery. Post-operative prophylaxis of ≤72 hours may be considered for anterior approach skull base procedures with intradural tumor resection or high flow spinal fluid leak anticipated. Agent: Amoxicillin/clavulanate or Trimethoprim/Sulfamethoxazole	Adults: **Cefuroxime** + **Metronidazole** *If history of MRSA infection or colonization:* + **Vancomycin** **Vancomycin** + **Aztreonam** + **Metronidazole**

OBSTETRICAL AND GYNECOLOGICAL PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Cesarean section	Enteric gram-negative bacilli Anaerobes Group B Streptococci Enterococci	Adult: Not in Labor**:** **Cefazolin** In Labor (contractions with cervical dilation or membrane rupture): **Cefazolin** + **Azithromycin** 500 mg IV *Re-dose cefazolin during c-section for patients when peri-partum hemorrhage reaches 1,500ml and patient continue to have bleeding. Patient could receive an addition second intra-operative dose 4-hours after the first re-dose if closure has not occurred yet. Redosing of azithromycin, clindamycin or gentamicin is not recommended for patients with blood loss.	Not in Labor: **Vancomycin** + **Gentamicin** 5 mg/kg IV x 1 (adjusted body weight) In Labor (contractions with cervical dilation or membrane rupture): **Vancomycin** + **Azithromycin** 500 mg IV + **Gentamicin** 5 mg/kg IV x 1 (adjusted body weight)
Vaginal hysterectomy **Abdominal hysterectomy** **Laproscopic or robotic hysterectomy** (including supracervical hysterectomy) **Urogynecology procedures including those involving mesh** **Enterocele repair**	Enteric gram-negative bacilli Anaerobes Group B Streptococci Enterococci	Adult: **Cefazolin** + **Metronidazole**	Adult: **Clindamycin** + **Gentamicin** 5 mg/kg IV x1 (if pregnant, use adjusted body weight; otherwise, use ideal body weight) OR **Clindamycin** + **Aztreonam**

OBSTETRICAL AND GYNECOLOGICAL PROCEDURES			
Hysterosalpingogram or Chromotubation (In patients with no history of PID, HSG can be performed without prophylactic antibiotics. If HSG shows dilated fallopian tubes, antibiotic prophylaxis should be given)	Chlamydia	No history of PID: No antibiotics History of PID: **Doxycycline** 100 mg PO x1 Dilated fallopian tubes: **Doxycycline** 100 mg PO BID x5 days	
MVA **Suction D&C procedures** **D&E procedures**	Anaerobes	Adult: **Doxycycline** 100 mg PO one hour before procedure OR **Azithromycin** 1 g prior to procedure	
Cerclage	Enteric gram-negative bacilli Anaerobes Group B Streptococci Enterococci	Planned: Prophylaxis Not Recommended Emergent: **Ampicillin-sulbactam**	*Emergent:* **Cefazolin** *Emergent:* **Clindamycin** + **Gentamicin** 5 mg/kg IV x 1 (if pregnant, use adjusted body weight; otherwise, use ideal body weight) OR **Clindamycin** + **Aztreonam**
Laparoscopy or laparotomy that is not accompanied by hysterectomy **Conization of cervix** **Vulvectomy** (simple) **Laser treatment to vulva or perineum** **Cystocele and rectocele repair** **Perineorraphy WITHOUT mesh**	None	Prophylaxis Not Recommended	Prophylaxis Not Recommended

NEUROSURGICAL PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Craniotomy **VP shunts & other prosthetic material** **Spine implantable devices**	*S. aureus* *S. epidermidis*	Adults: **Cefazolin**	Adults: **Vancomycin**
Discography (Intradiscal antimicrobial prophylaxis is not endorsed by the UMHS Pharmacy and Therapeutics Committee and should not be used.)	*S. aureus* *S. epidermidis*	Adults: **Cefazolin** If history of MRSA infection or colonization: + **Vancomycin**	Adults: **Vancomycin**
Skull base, anterior approach including transphenoidal surgery for pituitary tumors	*S. aureus* *S. epidermidis* gram-negative bacilli	Adults: **Ampicillin-sulbactam** If history of MRSA infection or colonization: + **Vancomycin** No post-operative prophylaxis for routine pituitary adenoma surgery. Post-operative prophylaxis of ≤72 hours may be considered for anterior approach skull base procedures with intradural tumor resection or high flow spinal fluid leak anticipated. Agent: Amoxicillin/clavulanate or Trimethoprim/Sulfamethoxazole	Adults: **Cefuroxime** + **Metronidazole** If history of MRSA infection or colonization: + **Vancomycin** **Vancomycin** + **Aztreonam** + **Metronidazole**
Skull base, lateral or posterior approach procedures	*S. aureus* *S. epidermidis*	Adults: **Ampicillin-sulbactam** OR **Cefuroxime**	Adults: **Clindamycin**

OPHTHALMIC PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Ophthalmic procedures	*S. aureus* *S. epidermidis* Streptococci enteric gram-negative bacilli *Pseudomonas* spp.	Minimal evidence supporting routine use of prophylactic antibiotics for ophthalmic surgery. Discretion advised regarding drug choice, duration, or route of administration.	**Gentamicin** OR **Tobramycin** OR **Ciprofloxacin** OR **Gatifloxacin** OR **Levofloxacin** OR **Neomycin-Gramicidin-Polymixin B** Administer multiple drops topically over 2-72 hrs

ORTHOPEDIC PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Total joint replacement (Arthroplasty) **Implantation of prosthetic material** (e.g., intramedullary nails, screw, plates, wires) **Hip fracture repair**	*S. aureus* *S. epidermidis*	Adults: *S. aureus* nasal screen: - Negative for MRSA - Positive for MSSA - Not performed and in the absence of history of MRSA carriage or infection **Cefazolin** *S. aureus* nasal screen: - Positive for MRSA - Not performed, but patient has history of MRSA carriage or infection **Daptomycin** + **Cefazolin** Antibiotic prophylaxis should be discontinued within 24 hours following sugery	Adults: **Daptomycin** *Alternative to daptomycin:* **Vancomycin**

ORTHOPEDIC PROCEDURES			
Open Fracture Repair (Includes upper and lower extremity open fractures) **NOTE: These are abbreviated recommendations. Please see full guideline on antimicrobial stewardship website (under 'Bone and Joint Infections' header):** **https://antimicrobialstewardship.med.umich.edu/guidelines/adult**	*S. aureus* *Streptococcus* *gram-negative rods*	Adults: *Type I or II open fracture:* **Cefazolin** Extensive soil exposure (e.g., farming accident) + **Metronidazole** *Type III open fracture:* **Ceftriaxone** Extensive soil exposure (e.g., farming accident) + **Metronidazole** Antibiotic prophylaxis should be provided for 24 hours.	Adults: *Type I or II open fracture* **Vancomycin** *Type III open fracture* **Vancomycin** + **Aztreonam** Antibiotic prophylaxis should be provided for 24 hours.

ORTHOPEDIC PROCEDURES			
Spinal procedure, with or without instrumentation	*S. aureus* *Coagulase negative staphylococci* *Gram negative bacilli*	Adults: *S. aureus* nasal screen: - Negative for MRSA - Positive for MSSA - Not performed and in the absence of history of MRSA carriage or infection **Cefazolin** *S. aureus* nasal screen: - Positive for MRSA - Not performed, but patient has history of MRSA carriage or infection **Daptomycin** + **Cefazolin** Antibiotic prophylaxis should be discontinued within 24 hours following sugery	Adults: **Daptomycin** *Alternative to daptomycin:* **Vancomycin**
Sports Medicine (Orthopaedic Surgeries) **Implantation of interference screws, suture anchors, permanent sutures, etc.**	*S. aureus* *Streptococcus*	Adults: **Cefazolin** History of MRSA infection or colonization: + **Daptomycin**	Adults: **Daptomycin**

REFERENCES
1. Bratzler DW, et al. Am J Health-Syst Pharm 2013; 70:195-283.
2. Schweizer ML et al. JAMA 2015; 313:2162-2171.
3. Rodriguez L et al. J Trauma Acute Care Surg 2013;77:400-408.

PLASTIC SURGERY PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Breast reconstruction (without implants), **cosmetic procedures** (excluding blepharoplasty), **large hand dissections*, phalloplasty, vaginoplasty, and general reconstructive procedures involving medium/large flaps or tissue expanders** *Open fractures: see Orthopedic Surgery guidelines	*S. aureus* *S. epidermidis*	Adults: **Cefazolin**	Adults: **Clindamycin**
Breast procedures with implants	*S. aureus* *S. epidermidis*	Adults: **Cefazolin** If history of MRSA infection or colonization: + **Daptomycin**	Adults: **Daptomycin**
Cleft lip and **palate repair,** or **facial procedures** that transect **oral, nasal (see rhinoplasty below), or pharyngeal mucosa**	Oral anaerobes enteric gram-negative bacilli *S. aureus* *S. epidermidis* viridans streptococci	Adults: **Ampicillin-sulbactam**	Adults: **Cefazolin** + **Metronidazole** **Levofloxacin** + **Metronidazole**

PLASTIC SURGERY PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Rhinoplasty/Septorhinoplasty	*S. aureus* *S. epidermidis* *C. acnes*	NOTE: Some experts do not recommend prophylaxis for all procedures. Prophylaxis may be considered based on complexity of procedure and comorbidities of patient. Adults: **Ampicillin-sulbactam** OR **Cefazolin**	Adults: **Clindamycin**

RADIOLOGY PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Any ARTERIAL OR VENOUS intervention that involves the hepatobiliary or reproductive systems. • Hepatic arterial embo for hepatic trauma, suspected hemobilia, Y-90 treatment (not mapping), TACE, or bland embo. • Any hepatic arterial embo in a transplanted liver or any liver at high-risk for ischemia (occluded/stenotic portal vein, etc.). • TIPS/DIPS Placement • BRTO/BATO/PARTO/CARTO • Portal vein embolization • Uterine artery embo • IVC filter retrieval with bowel penetration **Biliary Interventions** • PTC • Biliary drain check/change • Biliary stent placement • Cholecystostomy tube placement • Cholecystostomy tube check/change • Gallbladder cryo- and chemical ablation • Biliary endoscopy **Thermal ablation of the liver**	Skin Flora: *S. aureus, S. epidermis* GI Flora: Enteric gram-negative bacilli, anaerobes	Adults: **Cefuroxime** + **Metronidazole** OR **Cefoxitin** Patients with known multi-drug resistant (MDR) bacteria colonization: Consider discussing with ID regarding tailored IV antibiotic prophylaxis.	Adults: **Clindamycin** + **Gentamicin** *Alternative if any allergy to gentamicin **OR** SCr 2 mg/dL **OR** CrCl < 40 mL/min:* **Clindamycin** + **Aztreonam**
Any ARTERIAL EMBOLIZATION involving the GU system	Skin Flora: *S. aureus, S. epidermis* GU Flora: *E. coli, Proteus, Klebsiella*	Prophylaxis of genitourinary procedures that enter the urinary tract should include coverage of likely	Adults: **Clindamycin** ± **Gentamicin**

RADIOLOGY PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
• Prostate artery embolization • Renal embolization with > 70% of the kidney expected to be embolized. **GU Interventions** • PCN placement or check/change/conversion • NUS placement or check/change/conversion • UDC placement or check/change/conversion **Thermal ablation of the kidney with treatment zone including the collecting system** **Suprapubic Tube Placement**		pathogens as well as organisms from the *most recent urologic source* (urine culture, stone culture, etc.) within the preceding year, regardless of prior antimicrobial treatment. If the most recent positive urologic source culture was > 1 year ago, standard prophylaxis should be utilized Adults: **Cefuroxime** OR **Cefoxitin** Patients with known multi-drug resistant (MDR) bacteria colonization: Consider discussing with ID regarding tailored IV antibiotic prophylaxis.	*Alternative if any allergy to gentamicin **OR** SCr 2 mg/dL **OR** CrCl < 40 mL/min:* **Clindamycin** + **Aztreonam**
Renal Arterial Embolization with < 70% of the kidney expected to be embolized	Skin Flora: *S. aureus, S. epidermis* GU Flora: *E. coli, Proteus, Klebsiella*	**NOT USUALLY INDICATED**; consider in the following instances: • Immunosuppressed patient • Plan to leave sheath in place Adults: **Cefuroxime** OR **Cefoxitin**	Adults: **Clindamycin** ± **Gentamicin** *Alternative if any allergy to gentamicin **OR** SCr 2 mg/dL **OR** CrCl < 40 mL/min:* **Clindamycin** + **Aztreonam**

RADIOLOGY PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Any ARTERIAL, VENOUS, OR DIALYSIS angio or intervention with plan for covered stent placement, overnight lysis, or plan to leave sheath in place. • Any of these interventions that may result in placement of a covered stent, overnight lysis, or plan to leave the sheath in place. • Distal splenic artery embolization for trauma or aneurysm with > 70% of the spleen expected to be embolized • Partial Splenic Embolization for hypersplenism • AVF or AVG Declots **Lung Ablation** **PleurX Placement**	Skin Flora: *S. aureus, S. epidermis*	Adults: **Cefazolin** *If history of MRSA infection or colonization:* + **Daptomycin**	Adults: **Daptomycin**
Any ARTERIAL, VENOUS, OR DIALYSIS angio or intervention with NO plan for covered stent placement, overnight lysis, or plan to leave sheath in place. • Arterial, venous, or dialysis surveillance, recanalization, angioplasty, and bare metal stent placement.	Skin Flora: *S. aureus, S. epidermis*	**NOT USUALLY INDICATED;** consider in the following instances: • Immunosuppressed • Plan to place covered stent, overnight lysis, or plan to leave sheath in place	Adults: **Daptomycin**

RADIOLOGY PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
• Any arterial embolization not involving a solid organ. Includes bronchial artery embo, GDA embo, Y-90 mapping (not treatment), GI bleed embo (not if hemobilia is suspected), bariatric embo, genicular artery embo, or other visceral artery embo. • Proximal splenic artery embo • Distal splenic artery embo or splenic artery aneurysm embo with < 70% of the spleen expected to be embolized • IVC filter placement and IVC filter retrievals with NO bowel penetration. • Superficial Vein Ablation / Sclerotherapy • TIPS/DIPS check (no stent extension or relining). • Embo or sclero of vascular anomaly not involving the hepatobiliary, splenic, renal, GI, or GU systems. **Line placements and exchanges** **Thermal ablations of bone or kidney (treatment zone cannot include the collecting system)** **Gastrostomy/gastrojejunostomy placement/exchange, Jejunostomy exchanges** **Paracentesis, Thoracentesis** **Pain interventions** not requiring needle passage through bowel, liver, or kidneys.		Adults: **Cefazolin** *If history of MRSA infection or colonization:* + **Daptomycin**	
Embolization or sclerosis of vascular anomalies involving the hepatobiliary, splenic, renal, GI, or GU systems	Variable	Physician choice based on end-organ involvement	
Biopsy	Variable	**Not usually indicated;** consider in the following instances: immunosuppressed, needle trajectory	

RADIOLOGY PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
		traversing colonized organ system/fluid collection. Physician choice based on biopsy location.	
Abscess drain placement in patients receiving antibiotics and drain placement into a presumed sterile fluid collection (pleural effusion, hematoma, lymphocele, seroma)	Variable	**Not usually indicated** consider in the following instances: immunosuppressed, needle trajectory traversing colonized organ system/fluid collection. Antibiotics must be infusing or have finished infusing within 1 hour of procedure start. Physician choice based on biopsy location and/or presumed organism.	

SOLID ORGAN TRANSPLANT PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Kidney transplant	*S. aureus* *S. epidermidis* enteric gram-negative bacilli	Adults: **Cefazolin**	Adults: **Daptomycin** + **Aztreonam**
Liver transplant	Enteric gram-negative bacilli (e.g., *E. coli*, *Klebsiella*) Enterococci S. *aureus* anaerobes (*Bacteroides, Clostridia*)	Adults: **Piperacillin-tazobactam**	Adults: **Daptomycin** + **Aztreonam**
Pancreas transplant and Pancreas-Kidney Transplant	Enteric gram-negative bacilli (e.g., *E. coli*, *Klebsiella*) *S. aureus* anaerobes (*Bacteroides*, *Clostridia*)	Adults: **Cefoxitin** + **Fluconazole**	Adults: **Daptomycin** + **Aztreonam**
Laparoscopic Living Donor Nephrectomy	Staphylococci enteric gram-negative bacilli	Adults: **Cefazolin**	Adults: **Daptomycin** + **Aztreonam**
Living Donor Liver	Enteric gram-negative bacilli (e.g., *E. coli*, *Klebsiella*) Enterococci *S. aureus* anaerobes (*Bacteroides*, *Clostridia*) common with stents, biliary obstruction	Adults: **Cefoxitin**	Adults: **Daptomycin** + **Aztreonam**
For **Heart Transplant** see Cardiothoracic guideline For **Lung Transplant** see Thoracic (non-cardiac) guideline			

THORACIC (NON-CARDIAC) PROCEDURES

Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Esophagectomy	*S. aureus* *S. epidermidis*	Adults: **Cefazolin**	Adults: **Daptomycin**
Lung transplant	*S. aureus* aerobic gram-negative bacilli	Adults: **Vancomycin** + **Cefepime** All antimicrobials should be discontinued 48 hours post-operatively or until cultures are available	Adults: **Vancomycin** + **Aztreonam** All antimicrobials should be discontinued 48 hours post-operatively or until cultures are available
Misc. thoracic procedures	*S. aureus* *S. epidermidis*	Adults: **Cefazolin**	Adults: **Daptomycin**

VASCULAR PROCEDURES			
Nature of operation	Likely pathogens	Recommended regimen[α]	Alternative regimen[α] See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis**
Open, aortic aneurysm repair **Aortic and peripheral aneurysm repair with endovascular stent graft** **Arterial bypass with prosthetic graft**	*S. aureus* *S. epidermidis*	Adults: **Cefazolin** If history of MRSA infection or colonization + **Daptomycin**	Adults: **Daptomycin**
AV grafts (with prosthetic) & **fistulas** (no prosthetic) **with skin flap for vein transposition**	*S. aureus* *S. epidermidis*	Adults: **Cefazolin** If history of MRSA infection or colonization + **Daptomycin**	Adults: **Daptomycin**
Carotid endarterectomy with prosthesis or patch	*S. aureus* *S. epidermidis*	Adults: **Cefazolin** If history of MRSA infection or colonization: +**Daptomycin**	Adults: **Daptomycin**
Carotid endarterectomy without prosthesis or patch	None	Not recommended	Not recommended

VASCULAR PROCEDURES			
Fistulas (no prosthetic) **without skin flaps for vein transposition** **Varicose vein ablation** (laser or radio frequency) **Vena cava filter placement**	None	Not recommended	Not recommended
Lower extremity amputation for ischemia **Phlebectomy of varicose veins, stripping of varicose veins, ligation of varicose veins** (e.g., Saphenous Vein stripping or ligation)	*S. aureus* *S. epidermidis* enteric gram-negative bacilli *Clostridia* spp.	Adults: **Cefazolin**	Adults: **Daptomycin**
Thromboendarterectomy without bypass **Arterial bypass with vein graft**	*S. aureus* *S. epidermidis*	Adults: **Cefazolin** If history of MRSA infection or colonization + **Daptomycin**	Adults: **Daptomycin**

PATIENTS > 50 kg (ADULT) PRE-OP AND INTRAOPERATIVE ANTIBIOTIC DOSING				
Antimicrobial	**Pre-operative dose**[λ] Pre-operative dose does not require adjustment for renal dysfunction	**Intraoperative re-dosing*** Omit second re-dose in those with CrCl < 50 mL/min or on hemodialysis	**IV push**	**Infusion**
Ampicillin	2 g	2 g every 2 hours for 2 re-doses	3-5 min[ν]	30 min[δ]
Ampicillin/sulbactam	3 g	3 g every 2 hours for 2 re-doses	3-5 min[ν]	30 min[δ]
Aztreonam	2 g	2 g every 4 hours for 2 re-doses	3-5 min[ν]	30 min[δ]
Cefazolin	2 g if < 120 kg, 3 g if ≥ 120 kg	2 g (3 g if ≥ 120 kg) every 4 hours for 2 re-doses	3-5 min[ν]	30 min[δ]
Cefuroxime	1.5 g	1.5 g every 4 hours for 2 re-doses	3-5 min[ν]	30 min[δ]
Cefoxitin	2 g	2 g every 2 hours for 2 re-doses	3-5 min[ν]	30 min[δ]
Cefepime	2 g	2 g every 4 hours for 2 re-doses	3-5 min[ν]	30 min[δ]
Clindamycin	900 mg	900 mg every 6 hours for 2 re-doses	Not Recommended	30 min[δ]
Daptomycin	500mg if < 125kg 750mg if > 125kg[ζ]	Not Recommended	2 min	30 min[δ]
Piperacillin/tazobactam	4.5 g	4.5 g every 2 hours for 2 re-doses	Not Recommended	30 min[δ]
Metronidazole	500 mg	Not Recommended	Not Recommended	30 min[δ]
Ceftriaxone	2 g	Not Recommended	3-5 min[ν]	30 min[δ]
Gentamicin	5 mg/kg[ε] (ideal body weight)	Not Recommended	Not Recommended	30 min - 60 min
Vancomycin	1 g if < 80 kg, 1.5 g if ≥ 80 kg	1 g (1.5 g if ≥ 80 kg) every 8 hours for 2 doses	Not Recommended	60 – 120 min
Levofloxacin	500 mg	Not Recommended	Not Recommended	60 min[δ]
Ciprofloxacin	400 mg	Not Recommended	Not Recommended	60 min[δ]
Fluconazole	400 mg	Not Recommended	Not Recommended	120 min[δ]

PATIENTS ≤50 kg (ADULT) PRE-OP AND INTRAOPERATIVE ANTIBIOTIC DOSING RECOMMENDATIONS[ξ,Σ]

Antibiotic Dose	Recommended Concentration	Infusion Time	Patient Weight in kg												Intraoperative Redosing Interval
			< 5	5-7.49	7.5-9.9	10-14.9	15-19.9	20-24.9	25-29.9	30-34.9	35-39.9	40-44.9	45-50	> 50	
Ampicillin 50 mg/kg max: 2000 mg	1 g/10 mL	Maximum of 200 mg/min	USE MG/KG DOSING	375 mg	500 mg	750 mg	1000 mg	1250 mg	1500 mg	1750 mg	2000 mg	2000 mg	2000 mg	REFER TO >50 kg DOSING RECOMMENDATIONS	*q2h x2 redoses*
Ampicillin-Sulbactam Max: 3000 mg	1 g/10 mL	Maximum of 200 mg/min		560 mg	750 mg	1125 mg	1500 mg	1875 mg	2250 mg	3000 mg	3000 mg	3000 mg	3000 mg		*q2h x2 redoses*
Aztreonam 30 mg/kg max: 2000 mg	1 g/10 mL	IVP 3-5 minutes		225 mg	300 mg	450 mg	600 mg	750 mg	750 mg	1000 mg	1000 mg	1000 mg	1500 mg		*q4h x2 redoses*
*Cefazolin 50 mg/kg max: 2000 mg	1 g/10 mL	IVP 3-5 minutes		375 mg	500 mg	750 mg	1000 mg	1250 mg	1500 mg	1750 mg	2000 mg	2000 mg	2000 mg		*q4h x2 redoses*
Cefepime 50 mg/kg max: 2000 mg	1 g/10 mL	3-5 minutes		375 mg	500 mg	750 mg	1000 mg	1250 mg	1500 mg	1750 mg	2000 mg	2000 mg	2000 mg		*q4h x2 redoses*
Cefoxitin 40 mg/kg max: 2000 mg	1 g/10 mL	3-5 minutes		300 mg	400 mg	600 mg	800 mg	1000 mg	1250 mg	1500 mg	1500 mg	2000 mg	2000 mg		*q2 x2 redoses*
Ceftriaxone 50 mg/kg max: 2000 mg	40 mg/mL	3-5 minutes		375 mg	500 mg	750 mg	1000 mg	1250 mg	1500 mg	1750 mg	2000 mg	2000 mg	2000 mg		*None*
Cefuroxime 50 mg/kg max: 1500 mg	1 g/10 mL	3-5 minutes		375 mg	500 mg	750 mg	1000 mg	1500 mg	1500 mg	1500 mg	1500 mg	1500 mg	1500 mg		*q4h x2 redoses*
Ciprofloxacin 10 mg/kg max: 400 mg	2 mg/mL	Minimum of 60 minutes		75 mg	100 mg	150 mg	200 mg	250 mg	300 mg	350 mg	400 mg	400 mg	400 mg		*None*

PATIENTS ≤50 kg (ADULT) PRE-OP AND INTRAOPERATIVE ANTIBIOTIC DOSING RECOMMENDATIONS[ξ,Σ]

Antibiotic Dose	Recommended Concentration	Infusion Time	Patient Weight in kg											Intraoperative Redosing Interval
			< 5	5-7.49	7.5-9.9	10-14.9	15-19.9	20-24.9	25-29.9	30-34.9	35-39.9	40-44.9	45-50	> 50	
Clindamycin 20 mg/kg max: 900 mg	20 mg/mL	Maximum of 30 mg/min	USE MG/KG DOSING	150 mg	200 mg	300 mg	400 mg	500 mg	600 mg	700 mg	800 mg	900 mg	900 mg	REFER TO >50 kg DOSING RECOMMENDATIONS	*Q6h x2 redoses*
Co-trimoxazole 5 mg/kg TMP max: 160 mg TMP	80 mg SMX: 16 mg TMP/mL 5 ml vial Max conc 1:10	Minimum of 30 minutes		37.5 mg	50 mg	75 mg	100 mg	125 mg	150 mg	160 mg	160 mg	160 mg	160 mg		*Q6h x2*
Daptomycin max: 500 mg	50 mg/mL	IVP 3-5 minutes		USE 6 MG/KG DOSING						350 mg	350 mg	350 mg	500 mg		*None*
Fluconazole 6 mg/kg max: 400 mg	2 mg/mL	Maximum of 200 mg/hour		40 mg	50 mg	75 mg	100 mg	135 mg	165 mg	200 mg	225 mg	250 mg	285 mg		*None*
Gentamicin/ Tobramycin 5 mg/kg max: 250 mg	10 mg/mL	Minimum of 30 minutes		USE MG/KG DOSING If patient is receiving systemic aminoglycoside therapy prior to surgery, consult OR pharmacist to determine appropriate timing and necessity of peri-operative dosing											*None*
Levofloxacin 10 mg/kg max: 750 mg	5 mg/mL	Minimum of 60 minutes		75 mg	100 mg	150 mg	200 mg	250 mg	300 mg	350 mg	400 mg	450 mg	500 mg		*None*
Metronidazole 15 mg/kg (dosing rounded to be divisible by 5) max: 500 mg	5 mg/mL	Minimum of 30 minutes		100 mg	150 mg	200 mg	300 mg	375 mg	450 mg	500 mg	500 mg	500 mg	500 mg		*None*
Piperacillin/ Tazobactam 100 mg/kg based on Piperacillin max: 3375 mg	100 mg/mL	Minimum of 30 minutes		840 mg	1125 mg	1690 mg	2250 mg	2800 mg	3375 mg	3375 mg	3375 mg	3375 mg	3375 mg		*Q2h x2 redoses*
Vancomycin 15 mg/kg max: 1000 mg	1 g/100 mL	Minimum of 60 minutes		USE MG/KG DOSING If patient is receiving systemic vancomycin therapy prior to surgery, consult OR pharmacist to determine appropriate timing and necessity of peri-operative dosing											*Q8h x2 redoses*

IV. FOOTNOTES

[2] Low-risk allergies include: pruritus without rash, remote (>10 years) unknown reaction, patient denies allergy but is on record, mild/severe rash with no other symptoms (see exceptions in 'severe non-IgE reactions below), urticaria/hives with no other symptomsSee β-lactam allergy evaluation and empiric guidance for further information.

[3] High-risk allergies include: respiratory symptoms, angioedema, cardiovascular symptomssyncope/passing out, arrhythmia), anaphylaxis. **If a patient has a high-risk allergy to penicillins, they may receive cefazolin, cefoxitin, or cefuroxime.**. See β-lactam allergy evaluation and empiric guidance for further information.

[4] Severe non-IgE reactions that are contraindications to further beta-lactam use (**except aztreonam, which can be used unless the reaction was to ceftazidime or cefiderocol**) unless approved by Allergy: organ damage (kidney, liver), drug-induced immune-mediated anemia/thrombocytopenia/leukopenia, rash with mucosal lesions (Stevens Johnson Syndrome/Toxic Epidermal Necrosis), rash with pustules (acute generalized exanthematous pustulosis), rash with eosinophils and organ injury (DRESS – drug rash eosinophilia and systemic symptoms), rash with joint pain, fever, and myalgia (Serum Sickness). See β-lactam allergy evaluation and empiric guidance for further information.

[5] Beta-lactam antibiotics include the following antibiotic classes: penicillins, cephalosporins, carbapenems, aztreonam

α Refer to Pre-op and Intraoperative Antibiotic Dosing Recommendations Guidelines

- Guideline includes infusion duration and time to intra-operative redosing
- All prophylactic antimicrobials should be discontinued after the intra-operative period, unless otherwise specified
- Adult patients < 50 kg should receive antibiotic dosing based on ≤ 50 kg guideline
- Pediatrics patients > 50 kg should receive antibiotic dosing based on > 50 kg guideline
- Use mg/kg dosing for patients < 5 kg
- SIP operative pediatrics procedures include: cardiac, ventriculoperitoneal shunts, and spinal surgery

π Adapted from Clinical Infectious Diseases 2004;38:1706-15 and Am J Health-Syst Pharm 2013;70.

γ Reconstituted dose injected directly into vein or via running IV fluids (only if IV piggyback not available).

δ Intermittent IV infusion.

ε Gentamicin dose should be based on ideal body weight unless actual body weight is less than ideal body weight. Use alternative to gentamicin in adult patients if SCr ≥ 2 mg/dL or CrCL < 40 mL/min, unless otherwise specified.

ζ Daptomycin dosing is based on actual body weight

λ Infusion Timing:

- Infusions should begin 15-60 minutes prior to incision for all antimicrobial agents unless listed below:
  - Levofloxacin, ciprofloxacin, vancomycin, gentamicin, azithromycin, and fluconazole, which should begin 45-90 minutes prior to incision.
- If pre-operative antibiotics have already been administered but incision has been delayed more than 60 minutes beyond the maximum dosing window (i.e., beta-lactam antibiotics started >2 hours prior to incision), consider pre-operative re-dosing for all antibiotics except vancomycin and aminoglycosides. For vancomycin and aminoglycosides, please contact pharmacy to determine if re-dosing is appropriate.

ξ Patients receiving systemic antibiotics prior to procedure should still receive the standard pre-operative antimicrobial prophylaxis with appropriate timing of administration as outlined in the UMHS surgical antimicrobial prophylaxis guidelines. Given the risk of nephrotoxicity, in patients receiving vancomycin or aminoglycosides prior to procedure who need these agents for pre-operative antimicrobial prophylaxis, please consult pharmacy to see if treatment doses can be rescheduled such that administration begins 45-90 minutes prior to incision.

Σ In patients with known colonization or infection with drug-resistant pathogens, the standard pre-operative antimicrobial prophylaxis should generally still be administered unless otherwise specified in procedure-specific guidelines. Please contact the ID approval pager (adults: 30780; pediatrics: 36149) to discuss the case if concerned about: history of a multidrug-resistant organism (e.g., ESBL-producing organism, carbapenem-resistant Enterobacteriaceae, etc.) at the site of the procedure in the previous year OR history of MRSA colonization in the previous year.

Ω Deviations may be appropriate for staged procedures. For example, in TMJ replacement cases ampicillin/sulbactam prior to the intra-oral portion of the procedure followed by cefazolin prior to the neck incision.

Last revised: 04/2026	**Antimicrobial Stewardship Subcommittee:** 04/2026	**P&T:** 05/2026

References

	Citation
1	Milstone A. et al. Timing of Preoperative Antibiotic Prophylaxis: A modifiable risk factor for deep surgical site infections after pediatric spinal fusion. Pediatr Infect Dis J. 2008 Aug;27(8):704-8.
2	Kids' Campaign 2007 Pediatric Webcast Series: Reduce Surgical Complications and Prevent Surgical Site Infections. 5 Million lives Campaign/
3	Bratzler DW., Houch PM. Antimicrobial Prophylaxis for Surgery: An Advisory Statement from the National Surgical Infection Prevention Project. Am J Surg. 2005 Apr;189(4):395-404.
4	American Academy of Pediatrics: Antimicrobial Prophylaxis in Pediatric Surgical Patients. Pediatrics. 1984 Sep;74(3):437-9.
5	Bratzler DW, et al. Clinical practice guidelines for antimicrobial prophylaxis in surgery. Am J Health Syst Pharm. 2013 Feb 1;70(3):195-283.

Revision History

Date	Revision
08/2020	Revised OB/GYN section
10/2020	Revised Urology section
11/2020	Revised Gastrointestinal section
03/2021	Revised allergy wording, Orthopedic, Plastics, and Cardiothoracic sections
04/2021	Adjusted rhinoplasty/septorhinoplasty and vasectomy recommendations
12/2021	Revised Orthopedic section
03/2022	Revised dosing recommendations for patients ≤ 50 kg
10/2022	Added Radiology, updated allergy wording
11/2022	Revised pediatric vancomycin, gentamicin/tobramycin dosing
01/2023	Adjusted allergy wording, revised beta-lactam footnote
03/2023	Removed redosing recommendation for piperacillin-tazobactam during liver transplant
08/2023	Revised Gastrointestinal section
02/2024	Revised Cardiothoracic section
12/2024	Revised Orthopedic section for open fractures
03/2025	Replaced vancomycin with daptomycin throughout document; revised daptomycin dosing
12/2025	Added post-operative guidance for anterior approach skull base procedures.
04/2026	Revised cesarean section alternative prophylaxis guidance.

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source